Are subtle expressions too sparse to recognize?

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***Abstract*—As subtle emotions are slightly and involuntarily expressed, they need to be recorded by high-speed camera. Though this high frame-per-second rate allows better capture of subtle expressions, it typically generates a lot of redundant frames with rapid varying illumination and noise but without significant motions. The redundancy is analyzed and eliminated by Sparsity-Promoting Dynamic Mode Decomposition (DMDSP), which helps synthesize dynamically condensed sequences. Moreover, DMDSP can also visualize dynamics of subtle expressions in both temporal and spectral domains. As meaningful subtle expressions are temporarily sparse, DMDSP would be able to extract these meaningful dynamics and improve recognition rates of subtle expressions. The hypothesis is evaluated on CASME II, a database of spontaneous subtle facial expressions. Recognition performance measured by F1-score, recall and precision metrics showed a significant leap of improvement when DMDSP is used to preserve a small percentage of meaningful frames in sequences with temporally high sparsity levels.**

***Keywords*—Subtle emotion recognition; Dynamic mode decomposition; Temporal sparsity; Micro-expressions; CASME II;**

I. Introduction

Recognition of subtle emotions, or facial expressions that are often expressed voluntarily and unexpectedly in a very short duration remains a daunting task for both humans and machines. Frank et al. [1] points out that untrained people are able to recognize subtle emotions with an accuracy of 32% while trained people reached up to 47% accuracy for such tasks in a psychological experiment. Meanwhile, Le Ngo et al. [2] showed that in a similar experimental protocol a machine-trained classifier can achieve 44% accuracy rate. Moreover, the authors also analyzed classification performance according to different frame-lengths generated by temporal interpolation model (TIM) [3]. In brief, TIM takes original frames as input data to construct manifolds of facial expressions and then samples on the manifolds for a particular number of output frames. It is shown in [2] that modifying temporal dynamics of facial expressions can improve classification rates if the number of interpolated frames are small. However, when the number of frames are larger, classification rates are somewhat hampered due to over-interpolation. Therefore, an appropriate manipulation of the dynamics of subtle expressions is vital to boost the ability of classifiers in a recognition task.

The natural sparsity of dynamics in subtle expressions can be demonstrated by a sample function $f(t)$ in Fig. 1. As subtle expressions happen very swiftly, videos of such rapid motions need to be recorded at high sampling rate by a high-speed camera e.g. each video from the CASME II database [4] is recorded at 200 frames-per-second (fps). Though such frame rate helps fully capture dynamics of facial expressions e.g. $f(t) > 0$ in Figure 1, it also generates large dynamically insignificant samples e.g. $f(t) = 0$ in the same figure. In the context of subtle expressions, it means that only a fraction of recorded frames contain faces with subtle expressions while the rest of frames are possibly just neutral faces. Neutral faces may not provide sufficiently distinctive facial dynamics (as compared to identity information). Hence, the removal of these dynamically insignificant neutral-face samples would not hamper classification. Moreover, it could improve the classification rate as sequences become more dynamically dense (e.g. $f(t')$ in Figure 1), producing more informative features later on.

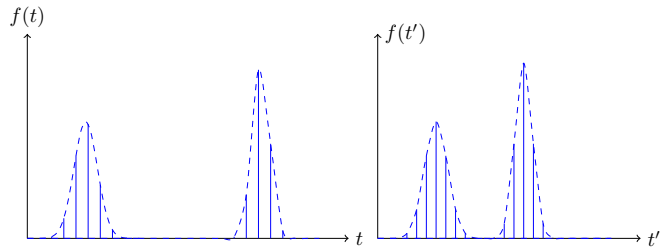


Fig. 1: Original signal *(left)*; Dynamically shorten signal *(right)*

In this paper, we propose the utilization of Sparse-Promoting Dynamic Mode Decomposition (DMDSP) [5] (described in Section II), to analyze the sparsity of temporal dynamics in subtle facial expressions. While standard Dynamic Mode Decomposition (DMD) deconstructs motions into several dynamic modes, DMDSP opt to find sparse combinations of these modes for reconstruction of the original dynamics. Section III shows the DMDSP analysis of subtle emotions in CASME II database and how redundant modes are exempted, resulting in the reduction of temporal dimension. Thereafter, the dynamically dense video samples are used in the evaluation of subtle emotion recognition. Further experimental details of the classifiers used, training and testing steps are presented in Section IV while Section V concludes this paper.

II. Sparsity-Promoting Dynamic Mode Decomposition

Dynamic Mode Decomposition (DMD) [6] is a data processing technique, capable of extracting coherent structures at a single temporal frequency or dynamic mode e.g. flows in fluid dynamics [6] and foreground motions in video processing [7]. Sparsity-Promoting Dynamic Mode Decomposition (DMDSP) [5] puts the decomposition under sparse constraints such that the least number of DMD modes are utilized for construction of original sequences. DMD and DMDSP were originally proposed for fluid dynamics imagery where a sequence of

"snapshots" form a data matrix equispaced across time. Formally, dynamics of pixel intensities from N sequential frames $\{\psi_0, \psi_1, \ldots, \psi_N\}$ are analyzed by DMD with the assumption that they evolve over regular temporal grids with a constant duration Δt between consecutive frames. From the pool of sequential frames, two clusters are formed as follows.

$$\Psi_0 := [\psi_0, \psi_1, \ldots, \psi_{N-1}] \qquad \Psi_1 := [\psi_1, \psi_2, \ldots, \psi_N]$$

In DMD assumption, pixel intensities of consecutive frames are governed by a linear time-invariant system A such that $\psi_{t+1} = A\psi_t$ for $t \in [0, \ldots, N-1]$. Hence,

$$\Psi_1 = [\psi_1, \psi_2, \ldots, \psi_N] = [A\psi_0, A\psi_1, \ldots, A\psi_{N-1}] = A\Psi_0 \tag{1}$$

For a rank-r matrix of the cluster of frames Ψ_0, DMD provides an optimal representation $F \in C^{rxr}$ in a proper orthogonal basis U as follows.

$$A \approx UFU^* \quad \Psi_0 = U\Sigma V^* \tag{2}$$

where U^* is a complex conjugate transpose of the basis U which is obtained from an economy-size singular value decomposition (SVD) of Ψ_0. The economy-size SVD is applicable as number of pixels M of each frame ψ_t are often much more than N, number of frames therefore $\Psi_0 \in C^{MxN}$ is a tall matrix. DMD models a linear time-invariant system $\Psi_1 = A\Psi_0$ by minimizing the following Frobenius norm.

$$\arg\min_A \|\Psi_1 - A\Psi_0\|_F^2 = \arg\min_F \|\Psi_1 - UF\Sigma V^*\| \tag{3}$$

By straightforward linear algebra, the optimal solution is achieved as:

$$F_{dmd} = U^*\Psi_1 V\Sigma^{-1} \tag{4}$$

As F_{dmd} is a rank-r matrix, it has a full set of linearly independent eigenvectors $\{y_1, \ldots, y_r\}$ and corresponding eigenvalues $\{\mu_1, \ldots, \mu_r\}$. Then, F_{dmd} can be expressed in diagonal form,

$$F_{dmd} = YD_\mu Z^* = [y_1 \quad \ldots \quad y_r] \begin{bmatrix} \mu_1 & & \\ & \ddots & \\ & & \mu_r \end{bmatrix} \begin{bmatrix} z_1^* \\ \vdots \\ z_r^* \end{bmatrix} \tag{5}$$

where $\{z_1, \ldots, z_r\}$ are eigenvectors of F_{dmd}^* which are biorthogonal to the $\{y_1, \ldots, y_r\}$, which means $Z^*Y = I$ and $F_{dmd}^t = YD_\mu^t Z^*$. As DMD assumes linear time-invariant relation between sequential frames, a frame ψ_t can be written in terms of an initial frame ψ_0 as below.

$$\psi_t = A^t\psi_0 \approx (UF_{dmd}U^*)^t\psi_0 = UYD_\mu^t Z^*U^*\psi_0 \tag{6}$$

Let $\Phi = UY$ as DMD modes and $A = Z^*U^*\psi_0$ as corresponding magnitudes, a DMD mode can be written as $\psi_t = \Phi D_\mu^t A$. With the amplitudes in D_α spanned by the Vandermonde matrix, V_{and} representing temporal evolution and Φ representing modes, the frame cluster Φ_0 is as follows:

$$\begin{aligned} [\psi_0 \ldots \psi_{N-1}] &\approx \Phi D_\alpha V_{and} \\ &= [\phi_0 \ldots \phi_r] \begin{bmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_r \end{bmatrix} \begin{bmatrix} 1 & \ldots & \mu_1^{N-1} \\ \vdots & \ddots & \vdots \\ 1 & & \mu_r^{N-1} \end{bmatrix} \end{aligned} \tag{7}$$

Determination of unknown amplitudes $\alpha := [\alpha_1 \ldots \alpha_r]^T$ depends on the solution of the following optimization problem,

$$\arg\min_\alpha \|J(\alpha)\| = \arg\min_\alpha \|\Psi_0 - \Phi D_\alpha V_{and}\|_F^2 \tag{8}$$

whereby the optimal DMD amplitudes can be obtained by,

$$\alpha_{dmd} = P^{-1}q = ((Y^*Y) \circ (V_{and}\bar{V}_{and}^*))^{-1} diag(V_{and}V\Sigma^*Y) \tag{9}$$

As DMD only analyzes data and does not apply any sparse constrain, the number of DMD modes as well as the number of reconstructed frames is equal to the number of frames. When sparse constraints are enforced, the DMDSP [5] takes on a constrained convex optimization problem:

$$\arg\min_\alpha \|J(\alpha)\| \quad \text{s.t.} \quad E^T\alpha = 0. \tag{10}$$

where E encodes the sparsity structure of the amplitude vector α. The desired sparsity structure is identified by iterating through a two-step minimization procedure solving for Eq. 8 augmented with an additional regularization term. Jovanovic et al. [5] showed that the optimal DMD amplitudes with a fixed sparsity structure can be determined from,

$$\alpha_{dmdsp} = [I \quad 0] \begin{bmatrix} P & E \\ E^T & 0 \end{bmatrix}^{-1} \begin{bmatrix} q \\ 0 \end{bmatrix} \tag{11}$$

III. Dynamics of Subtle Expressions

Subtle expressions, in the form of tiny movements of facial muscles, are difficult to capture through the tiny variation of intensities found at different locations. However, this temporal dynamics can be well summarized by the modes ϕ and amplitudes α of DMD (α_{dmd}) and DMDSP (α_{dmdsp}). As video data are complex and high-dimensional, r is equal to the number of frames N. There are no duplicate frames in the sequence since all frames are slightly different. This difference can be analyzed temporally by DMD modes ϕ and measured by their corresponding amplitudes α. Moreover, each DMD mode is extracted at a certain temporal frequency, enabling a subtle expression video to be analyzed in the spectral domain as well. Spectral analysis is able to reveal which spectra subtle emotions occur and whether they correspond to the general knowledge that subtle expression lasts between 1/15s and 1/25s.

A. Temporal Analysis

On a "happiness" sample from CASME II database (i.e "EP13_09" of Subject 17), DMD and DMDSP analysis are carried out to visualize amplitudes α over time t as shown in Fig. 2. Both α_{dmd} and α_{dmdsp} amplitudes reach their peaks at frame 21. While DMD has significant amplitudes at several other local maximas (e.g. 3, 70, 71, 80, 81, 94), while DMDSP with sparse constraint on amplitudes suppress all these local maximas. Moreover, Fig. 3a and 3b compare a neutral expression at frame 0 and the strongest expression at frame 21, respectively. By careful observation, viewers can spot a tiny but obvious movement at the subject's left mouth corner. The importance of the 21st mode is most obvious in the DMD modes ϕ_0 (Fig. 3c) and ϕ_{21} (Fig. 3d). The clarity of the reconstructed signal from mode ϕ_{21} is much better than that of mode ϕ_0, which ultimately explains the contribution of ϕ_{21} and how its amplitude α_{21} is much larger than α_0.

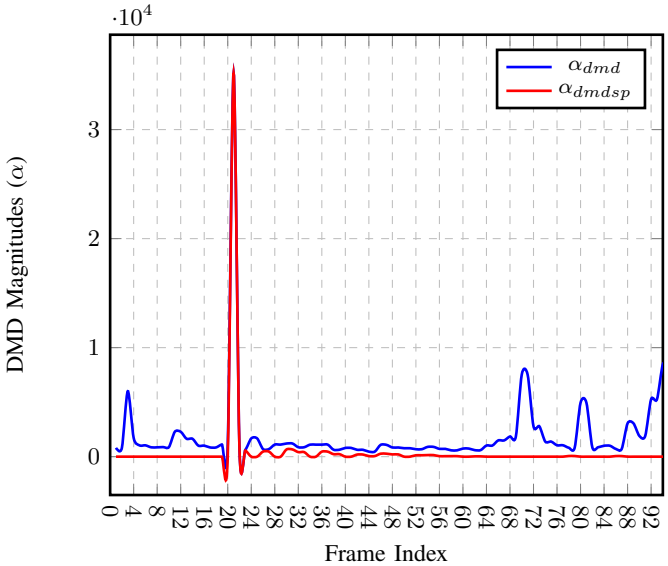
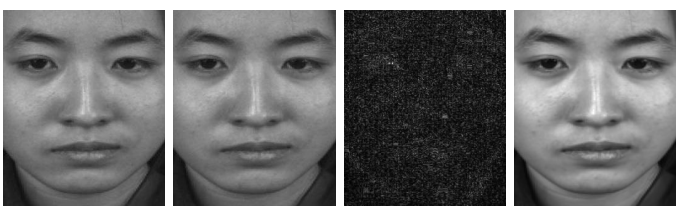


Fig. 2: DMD Magnitudes in Temporal Analysis



(a) ψ_0 (b) ψ_t (c) ϕ_0 (d) ϕ_{21}

Fig. 3: Original frames at index 0 and 21 (a)-(b); Signals reconstructed from modes 0 and 21 (c)-(d)

B. Spectral Analysis

The eigenvalues μ of D_μ from Eqn. 5 relate the DMD or DMDSP modes to single temporal frequencies [6], which are written in Fourier frequency modes as,

$$f_i = \frac{\log(\mu_i)}{2\pi\Delta t} \tag{12}$$

where Δt is the duration between two consecutive frames, i.e. $\frac{1}{200}s$ for CASME II videos. Fig. 4 shows plots of α_{dmd} and α_{dmdsp} over the bandwidth 1-100 Hz (no DC component)[1] on the same video sample. In Fig. 4, the DMD amplitudes have several spikes between 30 and 60 Hz, while DMDSP showed that most dominant amplitudes lie in frequency range 1-25 Hz. Subtle expressions typically last in a duration between $\frac{1}{25}s$ and $\frac{1}{15}s$ (or 15-25 Hz) while normal expressions last longer than $\frac{1}{15}$ second (or 1-15 Hz). Thus, DMDSP performs better than DMD in selecting the appropriate expression bandwidth while DMD somewhat captures the prominent frequency modes indiscriminately. To demonstrate the superiority of DMDSP over DMD in the extraction of meaningful dynamics, we visualize the modes ϕ at frequencies with maximum amplitudes i.e.

[1]With a sampling rate of 200 Hz, only dynamics with temporal frequencies less than 100 Hz could be captured based on Nyquist sampling theorem.

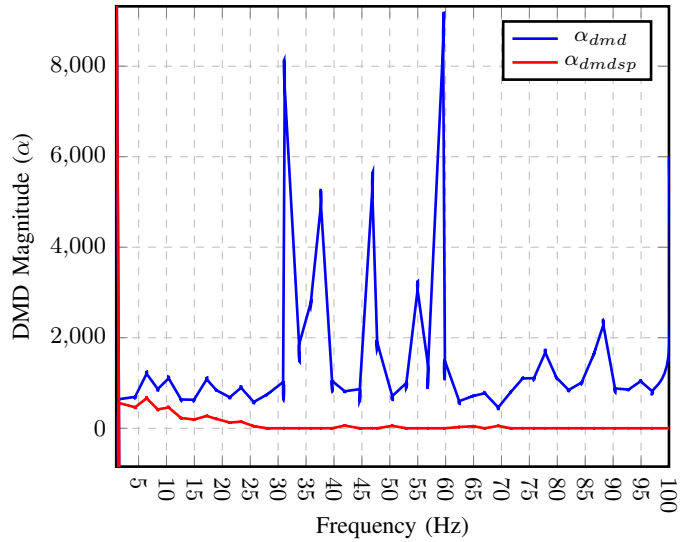
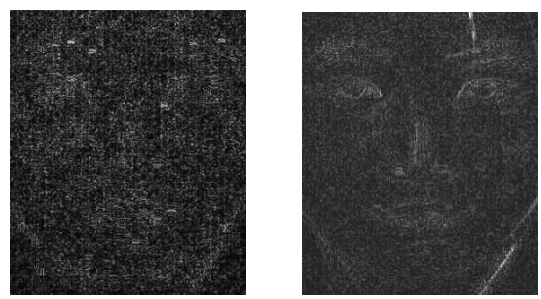


Fig. 4: DMD Magnitudes in Spectral Analysis



(a) DMD mode at 59.55 Hz (b) DMDSP mode at 6.44 Hz

Fig. 5: Spectral analysis of the peak amplitude modes from DMD and DMDSP

DMD α_{dmd} at 59.55 Hz (Fig. 5a) and DMDSP α_{dmdsp} at 6.44 Hz (Fig. 5b). The DMDSP mode at 6.44 Hz reflects more meaningful dynamics than the DMD mode at 59.55 Hz which suffers greatly from noise. While DMD has most amplitudes on the noisy modes, DMDSP highlights the more dynamically significant modes.

C. Sparsity Analysis

DMDSP puts sparsity constraints on amplitudes to limit the number of modes ϕ with non-zero amplitudes. These modes are reserved while the rest are eliminated. Hence, the reconstructed sequences have fewer frames with dynamically significant modes while frames with insignificant dynamics are removed. This essentially makes a sequence more dynamically compact than the temporally sparse signals in the original sequence (Fig. 2). We observe that the sparser the subtle emotions in a sequence, the fewer the number of useful frames are to be retained. Therefore, sparsity level is inversely proportional to a percentage of preserved frames, e.g. DMDSP keeps 45% of total frames in the previous example. The

optimal sparsity level across the dataset can be found through experiments.

IV. Experimental Results & Discussion

Experiments were carried out to investigate the following questions: *a*) Does DMDSP help improve the capability of recognizing subtle emotions? *b*) What percentage (or sparsity level) of the original video sequence are the most optimal for producing the best recognition rate?

A. Experiment Setup

After the video sequences processed by DMDSP to extract the dynamically significant frames, the corresponding synthesized sequence are used for classification of spontaneous subtle expressions. Following experimental protocol suggested by Le Ngo et al. [2], we choose CASME II [4], a spontaneous subtle expression database, which is collected, processed and annotated by psychologists in a systematic way. Though there are seven classes of subtle emotions in the database, only 5 classes ('happy', 'disgust', 'repression', 'tense' and 'surprise') are learnt by classifiers while 'fear' and 'sadness' are ignored due to limited numbers of samples. The recognition task includes the followings steps:

- **Pre-processing:** DMDSP synthesizes dynamically significant sequences from original temporally redundant sequences.
- **Feature Extraction:** Local Binary Pattern Three Orthogonal Planes (LBP-TOP) extracts features from the synthesized sequences.
- **Classifiers:** Support Vector Machine (SVM) [8] with Radial Basis Function and Linear Discriminant Analysis (LDA) [9] [10] methods are utilized for classification.

Percentages of reserved frames are varied from reserves 45% up 100% of original frame-lengths. As the shortest video sample of CASME II has 23 frames, we fix the lower bound at 45% to leave sufficient frames for feature extraction (so that there are at least 10 frames left thereafter). We used the best possible configuration[2] of $\text{LBPTOP}_{4,4,4,1,1,4}$ as suggested by Yan et al. in the original paper [4]. SVM classifier is set with regularization parameter $C = 10000$ using Radial Basis Function (RBF) kernel. Leave-one-subject-out (LOSO) cross-validation as suggested by Le Ngo [2] is used to ensure independence of subtle emotion recognition from subject identities during evaluation. In each fold, all samples from a subject is selected for testing while the remaining samples are used for training (CASME II has 26 subjects, hence 26 folds altogether). We use F1-score, precision and recall rates as performance metrics instead of average accuracy as CASME II is highly imbalanced with respect to the number of samples in each emotion class [2].

B. Experimental Results

Evaluation results in Table I compares the baseline results (no DMDSP processing) against results with different sparsity levels. For a better picture of the marked improvement when the sparsity of subtle facial dynamics are addressed,

[2]The first three numbers are the number of neighbors in the three planes, while the last three numbers are the radii of the three planes.

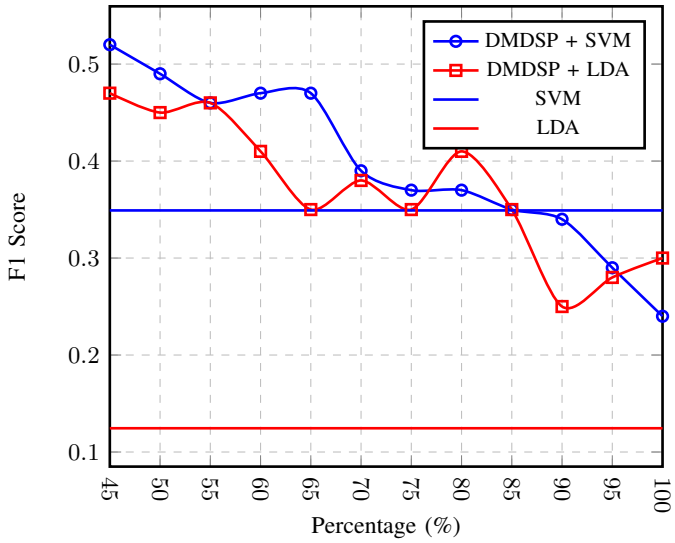


Fig. 6: F1-score versus percentage of preserved frames

TABLE I: F1-score, precision and recall results

	DMDSP + SVM			DMDSP + LDA		
Per	F1	Rec	Pre	F1	Rec	Pre
Baseline	0.35	0.34	0.35	0.12	0.21	0.09
45%	0.52	0.48	0.57	0.47	0.42	0.53
50%	0.49	0.46	0.53	0.45	0.42	0.48
55%	0.46	0.43	0.49	0.46	0.43	0.51
60%	0.47	0.43	0.50	0.41	0.41	0.41
65%	0.47	0.43	0.52	0.35	0.35	0.35
70%	0.39	0.35	0.44	0.38	0.33	0.43
75%	0.37	0.33	0.43	0.35	0.31	0.42
80%	0.37	0.34	0.41	0.41	0.37	0.46
85%	0.35	0.31	0.40	0.35	0.30	0.40
90%	0.34	0.31	0.38	0.25	0.28	0.22
95%	0.29	0.26	0.32	0.28	0.28	0.27
100%	0.24	0.22	0.26	0.30	0.27	0.33

the F1-scores of the proposed method are plotted in Fig. 6 against their corresponding baselines. Generally, when sparsity level increases (smaller percentage), recognition ability also improves. A stark improvement is observed with the LDA classifier, where DMDSP always results in better performance than its baseline result. Meanwhile, SVM performs better than its baseline when percentages are lower than 85%. Overall, the recognition rates peak at 45% of preserved frames where performance improved by $\approx 50\%$ for SVM and almost 3 times for the LDA classifier.

V. Conclusion

Subtle expressions are often dynamically sparse with many insignificant dynamic modes. DMDSP with sparse constraints eliminates these irrelevant modes and condenses the essential expression dynamically. Experimental results show that DMDSP is able to significantly improve subtle expression recognition under high sparsity levels where a low percentage of important frames are preserved. Moreover, subtle emotions are most recognizable when DMDSP preserves modes of 1-25 Hz, a known temporal range of frequencies for subtle and normal facial expressions.

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